Proving $A \subseteq B$

Let $A = \{x \in \mathbb{R} : x^2 < 5\}$ and let
$B = \{x \in \mathbb{R} : x \leqslant 4\}$

Prove that $A \subseteq B$

Proof. Let $x \in A$

Then $x^2 < 5$
Therefore $-\sqrt{5} < x < \sqrt{5}$
But $\sqrt{5} \leqslant 4$
Thus $x \leqslant 4$
Hence $x \in B$
So every element of A is an element of B, showing that $A \subseteq B$